

房. 一道题. 看这积分不等式的各种解法 $-M \leq f' \leq M$

例0. 设 $f(x)$ 在 $[a, b]$ 连续可导. $f(a)=0$. $|f'(x)| \leq M$. 证: $\left| \int_a^b f(x) dx \right| \leq \frac{M}{2} (b-a)^2$ $= M \cdot \frac{1}{2} (x-a)^2 \Big|_a^b = \frac{M(b-a)^2}{2}$

(证1): $\left| \int_a^b f(x) dx \right| = \left| \int_a^b [f(x) - f(a)] dx \right| = \left| \int_a^b (x-a) \cdot f'(x) dx \right| \leq \int_a^b (x-a) \cdot |f'(x)| dx \leq M \cdot \int_a^b (x-a) dx$

(证2): $\left| \int_a^b f(x) dx \right| = \left| \int_a^b [f(x) - f(a)] dx \right| = \left| \int_a^b \left(\int_a^x f'(t) dt \right) dx \right| \leq \int_a^b \left[\int_a^x |f'(t)| dt \right] dx \leq M \cdot \int_a^b (x-a) dx = \frac{M(b-a)^2}{2}$

(证3): $\left| \int_a^b f(x) dx \right| = \left| \int_a^b f(x) dx (x-b) \right| = \left| 0 - \int_a^b (x-b) f'(x) dx \right| \leq \int_a^b (b-x) |f'(x)| dx \leq M \cdot \int_a^b (b-x) dx$

$= M \cdot \int_a^b (x-a) dx = \frac{M(b-a)^2}{2}$ ✓

(证4): 即证: $-\frac{M}{2} (b-a)^2 \leq \int_a^b f(x) dx \leq \frac{M}{2} (b-a)^2$. 先证右半. 令 $F(x) = \int_a^x f(t) dt - \frac{M}{2} (x-a)^2$. $F(a)=0$

$F'(x) = f(x) - M(x-a) = [f(x) - f(a)] - M(x-a) = (x-a) \cdot [f'(x) - M] \leq 0 \Rightarrow F(x) \downarrow \Rightarrow F(b) \leq F(a) = 0$.

左半同理可证. 证毕!

(证5): 令 $F(x) = \int_a^x f(t) dt \Rightarrow |F(b)| = \left| \int_a^b f(t) dt \right| = \left| \int_a^b \frac{f''(t)}{2!} (t-a)^2 dt \right| \leq \frac{M}{2} (b-a)^2$ ✓

考法一. 将积分上限改为 x . 构造函数. 求导判断单调性. 总一分.

(该方法与《考题1. 微分不等式》的考法一 对应. 是最常规的方法!)

例1. 设 $f(x)$ 在 $[0,1]$ 连续可导. $0 < f(x) \leq 1$. $f(0)=0$ ⁽¹⁾ 证: $(\int_0^1 f(x) dx)^2 \geq \int_0^1 f^2(x) dx$. ⁽²⁾ ✓

(2) 并分析 $f(x)$ 为什么函数时. 该不等式恰好取到等号.

解: ⁽¹⁾ $F(x) = (\int_0^x f(t) dt)^2 - \int_0^x f^2(t) dt$. $F(0)=0$ (目的: 证 $F(1) \geq 0$) $\frac{1}{4} \geq \frac{1}{4}$

$$\Rightarrow F'(x) = 2 \int_0^x f(t) dt \cdot f(x) - f^2(x) = f(x) [2 \int_0^x f(t) dt - f(x)]$$

$$\text{令 } g(x) = 2 \int_0^x f(t) dt - f^2(x) \quad g(0)=0 \quad g'(x) = 2f(x) - 2f(x) \cdot f'(x) = 2f(x) [1 - f'(x)] \geq 0$$

$$\Rightarrow g(x) \uparrow \Rightarrow g(x) \geq 0 \Rightarrow F'(x) = f(x) \cdot g(x) \geq 0 \Rightarrow F(x) \uparrow \Rightarrow F(1) \geq F(0) = 0 \quad \text{证毕}$$

(2) 若取到等号. 即 $F(1)=0$. 此时 $F(x) \equiv 0 \Rightarrow F'(x) \equiv 0$. 又 $f(x)$ 在 $x \in (0,1]$ 有 $f(x) > 0$

$$\Rightarrow g(x) \equiv 0 \Rightarrow g'(x) \equiv 0 \Rightarrow 1 - f'(x) \equiv 0 \Rightarrow f'(x) \equiv 1 \Rightarrow f(x) = x + C. \text{ 又 } f(0)=0 \Rightarrow f(x)=x$$

例2. 设 $f(x)$ 在 $[a, b]$ 连续且递增, 证明: $\int_a^b x f(x) dx \geq \frac{a+b}{2} \int_a^b f(x) dx$ $x - \frac{a+b}{2}$

[解]: $F(x) = \int_a^x t f(t) dt - \frac{a+x}{2} \int_a^x f(t) dt$. $F(a) = 0$ (目的: $F(b) \geq 0$)

$$\Rightarrow F'(x) = \underbrace{x f(x)} - \frac{1}{2} \int_a^x f(t) dt - \frac{a+x}{2} f(x) = \frac{x-a}{2} f(x) - \frac{1}{2} \int_a^x f(t) dt \stackrel{\text{积分中值定理}}{=} \frac{x-a}{2} f(x) - \frac{x-a}{2} f(\xi)$$

$$= \frac{x-a}{2} [f(x) - f(\xi)]$$

其中 $a < \xi < x \Rightarrow F'(x) \geq 0 \Rightarrow F(x) \uparrow \Rightarrow F(b) \geq F(a) = 0$. 证毕!

[注]: 将条件中的“连续”弱化为“可积”. 则变限积分函数不能求导. 那应如何证明? 

[解]: 由于 $f(x) \uparrow \Rightarrow (x - \frac{a+b}{2}) \cdot [f(x) - f(\frac{a+b}{2})] \geq 0$. ✓

在 $[a, b]$ 积分: $\int_a^b (x - \frac{a+b}{2}) \cdot [f(x) - f(\frac{a+b}{2})] dx \geq 0$

$$\Rightarrow \int_a^b (x - \frac{a+b}{2}) f(x) dx - \int_a^b (x - \frac{a+b}{2}) f(\frac{a+b}{2}) dx \geq 0$$

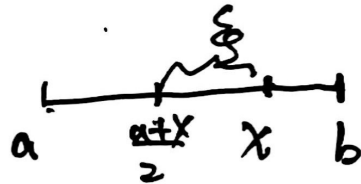
$$\Rightarrow \int_a^b x f(x) dx \geq \frac{a+b}{2} \int_a^b f(x) dx$$

$$\frac{1}{2} (x - \frac{a+b}{2})^2 \Big|_a^b$$

$$\geq \frac{1}{2} (\frac{b-a}{2})^2 - \frac{1}{2} (\frac{a-b}{2})^2$$

例3. (阿达玛不等式). $f(x)$ 在 $[a, b]$ 上满足 $f''(x) > 0$ 证: $f\left(\frac{a+b}{2}\right) < \frac{\int_a^b f(x) dx}{b-a} < \frac{f(a)+f(b)}{2}$

[解]: (1) 先证右边: $F(x) = (x-a)f\left(\frac{a+x}{2}\right) - \int_a^x f(t) dt$. $F(a) = 0$ (目的: $F(b) < 0$)



$$F'(x) = 1 \cdot f\left(\frac{a+x}{2}\right) + (x-a)f'\left(\frac{a+x}{2}\right) \cdot \frac{1}{2} - f(x) = \frac{a-x}{2}f'\left(\frac{a+x}{2}\right) + \frac{x-a}{2}f'\left(\frac{a+x}{2}\right) \quad \left(\frac{a+x}{2} < \xi < x\right)$$

$$= \frac{x-a}{2} [f'\left(\frac{a+x}{2}\right) - f'(\xi)]$$

由于 $f''(x) > 0 \Rightarrow f'(x)$ 严格递增! $\Rightarrow f'(\xi) > f'\left(\frac{a+x}{2}\right)$ (此时仅在 $x=a$ 时取等)

$\Rightarrow F(x)$ 严格单调递减 $\Rightarrow F(b) < F(a) = 0$. 证毕!

(2) $g(x) = \frac{f(a)+f(x)}{2} \cdot (x-a) - \int_a^x f(t) dt$. $g(a) = 0$ (目的: 证 $g(b) > 0$)

$$g'(x) = \frac{f'(x)}{2}(x-a) + \frac{f(a)+f(x)}{2} \cdot 1 - f(x) = \frac{x-a}{2}f'(x) + \frac{f(a)-f(x)}{2} = \frac{x-a}{2}f'(x) + \frac{a-x}{2}f'(a) \quad (a < \xi < x)$$

$$= \frac{x-a}{2} [f'(x) - f'(a)]$$

由于 $f''(x) > 0 \Rightarrow f'(x)$ 严格递增: $\Rightarrow g'(x) > 0$. (仅在 $x=a$ 时取等)

$\Rightarrow g(x)$ 严格递增 $\Rightarrow g(b) > g(a) = 0$. 证毕!

集题 (2022年). 设 $f(x)$ 在 $(-\infty, +\infty)$ 有三阶连续导数. 证明: $f''(x) \geq 0$ 的充要条件是“对不同的

实数 a, b , 均有 $f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx$ ✓. 互逆则反.

[解]: (1) 设 $f''(x) \geq 0$. 仿照上题证明. 可得 $f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx$. 略 ✓.

(2) 再设. 对 $\forall a \neq b$. 有 $f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx$. (我们的目的是证明 $f''(x) \geq 0$ 恒成立)

反证法: 假设 $\exists x_0 \in (a, b)$. s.t. $f''(x_0) < 0$. 由 $f''(x)$ 的局部保号性可得

$\Rightarrow \exists \delta > 0$. s.t. $x \in (x_0 - \delta, x_0 + \delta)$. 均有 $f''(x) < 0$. 仿照 (1) 中论证

对 $(x_0 - \delta, x_0 + \delta)$ 区间的任意两个点 x_1, x_2 . 均有 $f\left(\frac{x_1+x_2}{2}\right) > \frac{1}{x_2-x_1} \int_{x_1}^{x_2} f(x) dx$

(这与 $f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) dx$ 矛盾!)

☆☆☆

综上. 证毕!

考法 = 构造变限积分函数. 将积分问题转化为微分问题!

例4. 设 $f(x)$ 在 $[2, 4]$ 上二阶连续可导. $f(3)=0$. 证: $\exists \xi \in (2, 4)$. s.t. $f''(\xi) = 3 \int_2^4 f(x) dx$

[解]: 令 $F(x) = \int_3^x f(t) dt$. 本题变为: $F''(x)$ 在 $[2, 4]$ 连续, $F(3) = F'(3) = 0$. 证: $F''(\xi) = 3[F(4) - F(2)]$

$$F(4) = \cancel{F(3)} + \cancel{F'(3)(4-3)} + \frac{F''(3)}{2!}(4-3)^2 + \frac{F''(\xi_1)}{3!}(4-3)^3 \quad \text{--- ①} \Rightarrow \text{①} - \text{②} \Rightarrow F(4) - F(2) = \frac{F''(\xi_1) + F''(\xi_2)}{6}$$

$$F(2) = \cancel{F(3)} + \cancel{F'(3)(2-3)} + \frac{F''(3)}{2!}(2-3)^2 + \frac{F''(\xi_2)}{2!}(2-3)^3 \quad \text{--- ②}$$

介值定理 $\downarrow$ $\frac{F''(\xi)}{3}$ 证毕!

例5. 设 $f(x)$ 连续, $|f(x)| \leq M$, $\int_0^1 f(x) dx = 0$. 证: 对 $\forall a' \in [0, 1]$, 均有 $|\int_0^{a'} f(x) dx| \leq \frac{M}{8}$

[解]: 令 $F(x) = \int_0^x f(t) dt$. $F(0) = F(1) = 0$ 且 $F'(x)$ 连续, $|F'(x)| \leq M$ (目的: $|F(x)| \leq \frac{M}{8}$)

$$\begin{aligned} \int_0^1 F(x) dx &= \cancel{F(x)} + \cancel{F'(x)(-x)} + \frac{F''(\xi_1)}{2!}(-x)^2 \quad \text{--- ①} & \text{①} \cdot (1-x) + \text{②} \cdot x \\ \int_0^1 F(x) dx &= \cancel{F(x)} + \cancel{F'(x)(1-x)} + \frac{F''(\xi_2)}{2!}(1-x)^2 \quad \text{--- ②} & \Rightarrow 0 = \cancel{F(x)} + 0 + \frac{x \cdot (1-x)}{2} F''(\xi_1) + \frac{(1-x)^2 x}{2} F''(\xi_2) \end{aligned}$$

$$\Rightarrow |F(x)| \leq \frac{M}{2} \cdot x \cdot (1-x) \left[\cancel{x} + \cancel{(1-x)} \right] = \frac{M}{2} \cdot x(1-x) \leq \frac{M}{2} \cdot \left(\frac{1}{2}\right)^2 = \frac{M}{8} \quad \text{证毕!}$$

例6. 设 $f(x)$ 在 $[a, b]$ 连续可导. $|f'(x)| \leq M$. $f(a) = f(b) = 0$. 证: $|\int_a^b f(x) dx| \leq \frac{(b-a)^2}{4} M$

解: $\frac{1}{2} F(x) = \int_a^x f(t) dt$. $F(a) = F'(a) = F'(b) = 0$ 且 $|F''(x)| \leq M$. (证: $|f(b)| \leq \frac{M}{4} (b-a)^2$.)

$$F\left(\frac{a+b}{2}\right) = \cancel{F(a)} + \cancel{F'(a) \frac{b-a}{2}} + \frac{F''(\xi_1)}{2!} \left(\frac{b-a}{2}\right)^2 \quad \text{--- ①}$$

$$F\left(\frac{a+b}{2}\right) = \cancel{F(b)} + \cancel{F'(b) \frac{a-b}{2}} + \frac{F''(\xi_2)}{2!} \left(\frac{a-b}{2}\right)^2 \quad \text{--- ②}$$

$$\text{②} - \text{①} \Rightarrow 0 = F(b) + \frac{(b-a)^2}{8} [F''(\xi_2) - F''(\xi_1)]$$

$$\Rightarrow |F(b)| \leq \frac{(b-a)^2}{8} \cdot (\underline{M} + \underline{M}) = \frac{M}{4} (b-a)^2 \quad \checkmark$$

例7. (2000年). 设 $f(x)$ 在 $[0, \pi]$ 上连续. 且 $\int_0^\pi f(x) dx = \int_0^\pi f(x) \cdot \cos x dx = 0$.

(证) 在 $(0, \pi)$ 内, 必存在两个不同的点 ξ_1, ξ_2 , s.t. $f(\xi_1) = f(\xi_2) = 0$. (f 在 $(0, \pi)$ 内有 2 个零点)

[解]: $F(x) = \int_0^x f(t) dt$, $F'(x) = f(x)$, $F(0) = F(\pi) = 0$. $0 = \int_0^\pi f(x) \cdot \cos x dx = \int_0^\pi \cos x dF(x)$.

$$= 0 - \int_0^\pi F(x) \cdot (-\sin x) dx = 0 \Rightarrow \int_0^\pi F(x) \cdot \sin x dx = 0 \text{ 由积中. } \exists c \in (0, \pi)$$

$$\text{s.t. } (\pi - 0) \cdot F(c) \cdot \sin c = 0 \Rightarrow F(c) = 0 \text{ 即 } F(0) = F(c) = F(\pi) = 0. \text{ 由罗尔定理}$$

$$\exists \xi_1 \in (0, c), \xi_2 \in (c, \pi), \text{ s.t. } F'(\xi_1) = F'(\xi_2) = 0 \text{ 即 } f(\xi_1) = f(\xi_2) = 0. \text{ 证毕!}$$

[证]: 这是一大美题! 把条件改为 (1) $\int_0^1 f(x) dx = \int_0^1 x f(x) dx = 0$, (2) $\int_0^1 f(x) dx = \int_0^1 e^x f(x) dx = 0$, (3) $\int_0^\pi \sin x f(x) dx = \int_0^\pi \cos x f(x) dx = 0$ 其条件, 均能证出相应结论

$$(1) F(x) = \int_0^x f(t) dt, F(0) = F(1) = 0, 0 = \int_0^1 x f(x) dx = \int_0^1 x dF(x) = 0 - \int_0^1 F(x) dx, \text{ 由积中. } F(c) = 0 \Rightarrow F(0) = F(c) = F(1) = 0. \text{ (后略)}$$

$$(2) F(x) = \int_0^x f(t) dt, F(0) = F(1) = 0, 0 = \int_0^1 e^x f(x) dx = \int_0^1 e^x dF(x) = 0 - \int_0^1 F(x) e^x dx \Rightarrow F(c) \cdot e^c = 0 \Rightarrow F(0) = F(c) = F(1) = 0. \text{ 后略!}$$

$$(3) F(x) = \int_0^x f(t) dt, F(0) = 0, \text{ 但 } F(\pi) = ??? \text{ 故换一函数! } \text{令 } F(x) = \int_0^x (\sin t) f(t) dt, F(0) = F(\pi) = 0. \\ 0 = \int_0^\pi \cos x \cdot f(x) dx = \int_0^\pi \frac{\cos x}{\sin x} f(x) \sin x dx = \int_0^\pi \cot x dF(x) = 0 - \int_0^\pi F(x) \cdot (-\csc^2 x) dx = \int_0^\pi F(x) \cdot \frac{1}{\sin^2 x} dx. \text{ 由积中.}$$

$$(\pi - 0) \cdot F(c) \cdot \frac{1}{\sin^2 c} = 0 \Rightarrow F(c) = 0. \text{ 故 } F(0) = F(c) = F(\pi) = 0. \text{ 由罗尔, } F'(\xi_1) = F'(\xi_2) = 0. \text{ 即 } \sin \xi_1 > 0, \sin \xi_2 > 0 \Rightarrow f(\xi_1) = f(\xi_2) = 0 \vee$$

考法三. 利用中值定理. 证明积分不等式.



(一). 利用积分中值定理

例8. 设 $f(x)$ 在 $[0, 1]$ 连续. 单调递减. 证: $\int_0^a f(x) dx \geq a \int_0^1 f(x) dx$ (其中 $0 < a < 1$)

[解]: (本题方法很多!)

证法: $f(x) \geq 0$ 在 $(0, 1)$ 成立

(证一): 令 $F(x) = \int_0^x f(t) dt - x \int_0^1 f(x) dx \Rightarrow F(0) = F(1) = 0$. $F'(x) = f(x) - \int_0^1 f(x) dx = f(x) - f(\xi)$.

其中 $0 < \xi < 1$. 又因 $f(x) \downarrow \Rightarrow f(x)$ 在 $(0, \xi) \uparrow$, 在 $(\xi, 1) \downarrow$. 再结合 $F(0) = F(1) = 0 \Rightarrow f(x)$ 在 $(0, 1)$ 内 $F(x) \geq 0$. 即 $F(a) \geq 0$ 即 $\int_0^a f(x) dx \geq a \int_0^1 f(x) dx$. 证毕!

(证二). 使用积分中值定理: 等价于证明 $\int_0^a f(x) dx - a \int_0^1 f(x) dx \geq a \int_a^1 f(x) dx$

$$\Leftrightarrow (1-a) \int_0^a f(x) dx \geq a \int_a^1 f(x) dx$$

$$\Leftrightarrow (1-a)(a-\xi_1)f(\xi_1) \geq a(1-a)f(\xi_2)$$

又因 $f(x) \downarrow$. 只需证: $\xi_1 \leq \xi_2$ 即可! 而 $\xi_1 \in (0, a)$, $\xi_2 \in (a, 1)$. 故 $\xi_2 > \xi_1$. 证毕!

习题. (2005年). 设 $f(x), g(x)$ 在 $[0,1]$ 连续可导. $f(0)=0, f'(x)>0, g'(x)>0, (f \cdot g)'$
 证明: 对于 $\forall a \in [0,1]$, 均满足 $\int_0^a f'(x) \cdot g(x) dx + \int_0^1 f(x) g'(x) dx \geq f(a) \cdot g(1)$ ✓

[解]: 我们予证: $\int_0^a (f'g + fg') dx + \int_a^1 f \cdot g' dx \geq f(a)g(1)$ ✓

$$\Leftrightarrow (f(a)g(a) - f(0)g(0)) + \int_a^1 f g' dx \geq f(a)g(1) \quad \checkmark$$

$$\Leftrightarrow f(a) \cdot [g(a) - g(0)] + \int_a^1 f(x) g'(x) dx \geq 0 \quad \checkmark$$

$$\Leftrightarrow f(a) \int_0^a g'(x) dx + \int_a^1 f(x) g'(x) dx \geq 0 \quad \checkmark$$

$$\Leftrightarrow \int_a^1 \underbrace{g'(x)}_{>0} \cdot \underbrace{[f(x) - f(a)]}_{>0} dx \geq 0 \quad \checkmark$$

对于 $f'(x) > 0$, 且 $x \in (a,1) \Rightarrow f(x) > f(a) \Rightarrow f(x) - f(a) > 0$. 而被积函数 $g'(x)[f(x) - f(a)] \geq 0$

$\Rightarrow \int_a^1 g'(x) \cdot [f(x) - f(a)] dx \geq 0$ 成立. 故证毕!

(二). 利用拉格朗日中值定理.



例9. 设 $f(x)$ 在 $[a, b]$ 连续可导. $|f'(x)| \leq M$. 证明:

$$F(x) = \int_a^x f(t) dt.$$

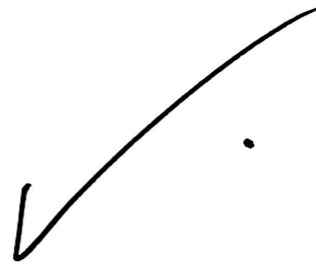
(1). 若 $f(a)=0$ 则 $|\int_a^b f(x) dx| \leq \frac{(b-a)^2}{2} M$; (2). 若 $f(a)=f(b)=0$ 则 $|\int_a^b f(x) dx| \leq \frac{(b-a)^2}{4} M$

[解]: (1) 证法. 略 $f(x) - f(a) = (x-a)f'(\xi)$

$$(2) \quad \left| \int_a^b f(x) dx \right| \leq \underbrace{\int_a^{\frac{a+b}{2}} |f(x)| dx}_{f(a)} + \underbrace{\int_{\frac{a+b}{2}}^b |f(x)| dx}_{f(b)} = \int_a^{\frac{a+b}{2}} |(x-a)f'(\xi_1)| dx + \int_{\frac{a+b}{2}}^b |(x-b)f'(\xi_2)| dx$$

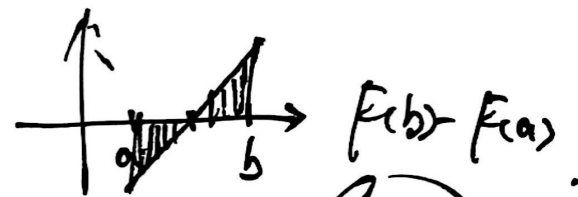
$$\leq M \cdot \left[\int_a^{\frac{a+b}{2}} (x-a) dx + \int_{\frac{a+b}{2}}^b (b-x) dx \right] = M \cdot \left[\frac{1}{2} (x-a)^2 \Big|_a^{\frac{a+b}{2}} - \frac{1}{2} (x-b)^2 \Big|_{\frac{a+b}{2}}^b \right]$$

$$= M \cdot \left[\frac{1}{2} \cdot \frac{(b-a)^2}{4} + \frac{1}{2} \cdot \left(0 - \frac{(b-a)^2}{4} \right) \right] = \frac{M}{4} \cdot (b-a)^2$$



(三). 利用泰勒中值定理

情形一: 已知 $|f''(x)| \leq M$. (泰勒展开以后, 放缩余项)



例10 设 $f(x)$ 在 $[a, b]$ 上连续可导. $f(\frac{a+b}{2}) = 0$, $|f''(x)| \leq M$. 证: $|\int_a^b f(x) dx| \leq \frac{(b-a)^3}{24} \cdot M$.

解:

$$f(x) = f(\frac{a+b}{2}) + f'(\frac{a+b}{2}) \left(x - \frac{a+b}{2}\right) + \frac{f''(\xi)}{2!} \left(x - \frac{a+b}{2}\right)^2$$

$$\int_a^b \left(x - \frac{a+b}{2}\right) dx = \frac{1}{2} \left(x - \frac{a+b}{2}\right)^2 \Big|_a^b = 0$$

$f(x)$ 在 $[a, b]$ 积分!

$$\left| \int_a^b f(x) dx \right| = 0 + 0 + \left| \int_a^b \frac{f''(\xi)}{2!} \left(x - \frac{a+b}{2}\right)^2 dx \right| \leq \frac{M}{2} \cdot \int_a^b \left(x - \frac{a+b}{2}\right)^2 dx$$

$$= \frac{M}{2} \cdot \frac{1}{3} \left(x - \frac{a+b}{2}\right)^3 \Big|_a^b = \frac{M}{6} \cdot \left[\left(\frac{b-a}{2}\right)^3 + \left(\frac{b-a}{2}\right)^3 \right] = \frac{M}{6} \cdot \frac{(b-a)^3}{4} \checkmark$$

另解: $f(x) = \int_{\frac{a+b}{2}}^x f(t) dt$.

情形二: 已知 $f(x)$ 恒正或恒负 (用 $f(x) \geq f(x_0) + f'(x_0)(x-x_0)$ 或 $f(x) \leq f(x_0) + f'(x_0)(x-x_0)$). (两边积分!)

例 11. (2018年). 设 $f(x)$ 在 $[0,1]$ 上可导, $\int_0^1 f(x) dx = 0$. 证明: 当 $f'(x) > 0$ 时, $f(\frac{1}{2}) < 0$.

解: 原题是选择题! $f(x) = f(\frac{1}{2}) + f'(\frac{1}{2})(x-\frac{1}{2}) + \frac{f''(\xi)}{2!}(x-\frac{1}{2})^2 \geq f(\frac{1}{2}) + f'(\frac{1}{2})(x-\frac{1}{2})$

两边在 $[0,1]$ 积分! $\int_0^1 f(x) dx \geq f(\frac{1}{2}) + 0 \Rightarrow f(\frac{1}{2}) \leq 0$

但由于 $f(x) \geq f(\frac{1}{2}) + f'(\frac{1}{2})(x-\frac{1}{2})$ 的等号仅在 $x=\frac{1}{2}$ 处取到!

$$\Rightarrow \int_0^1 f(x) dx > f(\frac{1}{2}) \Rightarrow f(\frac{1}{2}) < 0$$

类题 1. 设 $f(x)$ 在 $[0,1]$ 上可导, $f(\frac{1}{2}) = 1$, $f'(x) > 0$. 证: $\int_0^1 f(x) dx > 1$.

解: $f(x) > f(\frac{1}{2}) + f'(\frac{1}{2})(x-\frac{1}{2})$. 积分: $\int_0^1 f(x) dx > f(\frac{1}{2}) = 1$. 证毕!

类题 2. 设 $f'(x) \leq 0$. 证: $\int_0^1 f(x^2) dx \leq f(\frac{1}{3})$.

解: $f(x) \leq f(\frac{1}{3}) + f'(\frac{1}{3})(x-\frac{1}{3}) \Rightarrow f(x^2) \leq f(\frac{1}{3}) + f'(\frac{1}{3})(x^2-\frac{1}{3}) \Rightarrow \int_0^1 f(x^2) dx \leq f(\frac{1}{3})$

例12. (积分不等式). $f(x)$ 在 $[a, b]$ 上满足 $f''(x) > 0$ 证: $f\left(\frac{a+b}{2}\right) \leq \frac{\int_a^b f(x) dx}{b-a} \leq \frac{f(a)+f(b)}{2}$

【解】: $f(x) \geq f(x_0) + f'(x_0)(x-x_0)$ ★

(1) 先证右边. $f(x) \geq f\left(\frac{a+b}{2}\right) + f'\left(\frac{a+b}{2}\right)\left(x - \frac{a+b}{2}\right)$ 在 $[a, b]$ 积分

$$\int_a^b f(x) dx \geq (b-a) \cdot f\left(\frac{a+b}{2}\right) + 0 \Rightarrow \frac{\int_a^b f(x) dx}{b-a} \geq f\left(\frac{a+b}{2}\right) \quad d(a-x) = -dx$$

★★★ (2) 证左边. $f(x) = f(x) + f'(x)(a-x) + \frac{f''(\xi)}{2!}(a-x)^2 \geq f(x) + f'(x)(a-x)$

$$\text{两边积分: } (b-a)f(a) \geq \int_a^b f(x) dx + \int_a^b (a-x) df(x) = \int_a^b f(x) dx + (a-b)f(b) + \int_a^b f(x) dx$$

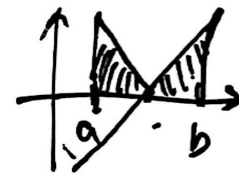
$$\Rightarrow (b-a) \cdot [f(a) + f(b)] \geq 2 \int_a^b f(x) dx \Rightarrow \frac{\int_a^b f(x) dx}{b-a} \leq \frac{f(a) + f(b)}{2} \quad \checkmark$$

综上证毕

考法三. 利用分部积分. 证明积分不等式.

例13. 设 $f(x)$ 在 $[a, b]$ 连续可导. $f(a)=0$. $|f'(x)| \leq M$. 证明: $\left| \int_a^b f(x) dx \right| \leq \frac{M}{2} (b-a)^2$

例14. 设 $f(x)$ 在 $[a, b]$ 连续可导. $f(a)=f(b)=0$. $|f'(x)| \leq M$. 证明: $\left| \int_a^b f(x) dx \right| \leq \frac{(b-a)^2}{4} M$.



解: $\left| \int_a^b f(x) dx \right| = \left| \int_a^b f(x) d\left(x - \frac{a+b}{2}\right) \right| = \left| 0 - \int_a^b \left(x - \frac{a+b}{2}\right) f'(x) dx \right| \leq \int_a^b \left|x - \frac{a+b}{2}\right| M dx$

$$= M \cdot 2 \int_{\frac{a+b}{2}}^b \left(x - \frac{a+b}{2}\right) dx = M \cdot \left(x - \frac{a+b}{2}\right)^2 \Big|_{\frac{a+b}{2}}^b = M \cdot \left(\frac{b-a}{2}\right)^2 = \frac{M}{4} (b-a)^2$$

(提示: $\int_a^b x \cdot h(x) dx \leq 0$)

例15. 设 $f(x), g(x)$ 在 $[a, b]$ 连续. $\int_a^x g(t) dt \leq \int_a^x f(t) dt$. $\int_a^b g(x) dx = \int_a^b f(x) dx$. 证明: $\int_a^b x f(x) dx \leq \int_a^b x g(x) dx$

解: 移项. 令 $h(x) = f(x) - g(x)$. 且 $H(x) = \int_a^x h(t) dt$. $H(a) = H(b) = 0 \Rightarrow H(x) \geq 0$

$$\int_a^b x h(x) dx = \int_a^b x dH(x) = 0 - \int_a^b H(x) dx \leq 0. \text{ 证毕!}$$

考法四. 利用积分与求和的统一. 证明积分不等式 (选学).

例16. 设 $f(x)$ 在 $[0,1]$ 连续. $|f(x)| \leq M$. 证明: $\left| \int_0^1 f(x) dx - \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \right| \leq \frac{M}{2n}$.

[解]: 左边 = $\left| \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x) dx - \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f\left(\frac{k}{n}\right) dx \right|$

$$= \left| \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left[f(x) - f\left(\frac{k}{n}\right) \right] dx \right|$$

$$\leq M \cdot \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(\frac{k}{n} - x \right) dx = M \cdot \sum_{k=1}^n \left(-\frac{1}{2} \right) \left(x - \frac{k}{n} \right)^2 \Big|_{\frac{k-1}{n}}^{\frac{k}{n}}$$

$$= M \cdot \sum_{k=1}^n \frac{1}{2} \cdot \frac{1}{n^2} = M \cdot n \cdot \frac{1}{2n^2} = \frac{M}{2n} \checkmark$$

考法五: 利用柯西不等式. 证明积不等式 (选学)

例1. 证明: 若 $f(x), g(x)$ 在 $[a, b]$ 连续, 则 $\int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx \geq \left[\int_a^b f(x)g(x) dx \right]^2$ (柯西不等式).

解: 证明方法极其多!!!

$$\text{令 } F(x) = \int_a^x f^2(t) dt \cdot \int_a^x g^2(t) dt - \left(\int_a^x f(t)g(t) dt \right)^2, \quad \underline{F(a)} = 0. \quad (\text{目的: 证 } F(b) \geq 0)$$

$$\underline{F'(x)} = \underline{f^2(x)} \cdot \int_a^x g^2(t) dt + \int_a^x f^2(t) dt \cdot \underline{g^2(x)} - 2 \int_a^x f(t)g(t) dt \cdot \underline{f(x)g(x)}:$$

$$= \int_a^x \left[\underline{f^2(x) \cdot g^2(t)} + \underline{f^2(t) \cdot g^2(x)} - 2 \cdot \underline{f(t)g(t) f(x)g(x)} \right] dt.$$

$$= \int_a^x \left[\underline{f(x)g(t) - f(t)g(x)} \right]^2 dt \geq 0 \Rightarrow F(x) \uparrow \Rightarrow \underline{F(b)} \geq F(a) = 0.$$

$$\text{即: } \int_a^b f^2 dx \cdot \int_a^b g^2 dx \geq \left(\int_a^b f \cdot g dx \right)^2. \quad \text{证毕!}$$

例18. 设 $f(x)$ 在 $[0,1]$ 连续, 且 $f(x) > 0$. 请利用柯西不等式证明: $\int_0^1 f(x) dx \cdot \int_0^1 \frac{1}{f(x)} dx \geq 1$

解: 左边 $\geq \left(\int_0^1 \left(\sqrt{f(x)} \cdot \frac{1}{\sqrt{f(x)}} \right) dx \right)^2 = 1 = \text{右边}$. 证毕!

类题: 证明: $\int_0^\pi x e^{\sin x} dx \cdot \int_0^\pi e^{-\cos x} dx \geq \frac{\pi^3}{4}$ $\int_0^\pi x f(\sin x) dx = \pi \cdot \int_0^{\frac{\pi}{2}} f(\sin x) dx$

解: 左边 $= \pi \cdot \int_0^{\frac{\pi}{2}} e^{\sin x} dx \cdot \int_0^{\frac{\pi}{2}} e^{-\cos x} dx$

$$\stackrel{\text{柯西}}{\geq} \pi \cdot \int_0^{\frac{\pi}{2}} e^{\sin x} dx \cdot \int_0^{\frac{\pi}{2}} e^{-\sin x} dx$$

$$\geq \pi \cdot \left(\int_0^{\frac{\pi}{2}} 1 dx \right)^2 = \pi \cdot \frac{\pi^2}{4} = \frac{\pi^3}{4} \checkmark$$

考法六. 逆用牛顿-莱布尼兹公式. 开放端积分区间. (逆学).

例19. 设 $f(x)$ 在 $[0, 2]$ 连续. 证明: $|f(2)| \leq \frac{1}{2} \left| \int_0^2 f(x) dx \right| + \int_0^2 |f'(x)| dx$ ★

解: 对于含有 $\int_a^b |f'(x)| dx$ 的积分不等式. 关键是制造出 $|f(x_2) - f(x_1)|$. 然后用绝对值不等式

回到本题: $|f(2)| = \left| \frac{1}{2} \int_0^2 f(x) dx \right| \xrightarrow{\text{积中}} |f(2)| - \frac{1}{2} \int_0^2 |f'(x)| dx = |f(2)| - |f'(c)| = |f(2) - f(c)| \leq |f(2) - f(c)|$

$$= \left| \int_c^2 f'(x) dx \right| \leq \int_c^2 |f'(x)| dx \leq \int_0^2 |f'(x)| dx$$

类题. 设 $f(x)$ 在 $[0, 1]$ 连续. 证: $\int_0^1 |f(x)| dx + \int_0^1 |f'(x)| dx \geq \max_{0 \leq x \leq 1} |f(x)|$ ★

解: 由积中. $\int_0^1 |f(x)| dx = |f(x_1)| \cdot 1, x_1 \in (0, 1)$. 又设 $\max_{0 \leq x \leq 1} |f(x)| = |f(x_2)|$

$$\Rightarrow \max |f(x)| - \int_0^1 |f(x)| dx = |f(x_2)| - |f(x_1)| \leq |f(x_2) - f(x_1)| = \left| \int_{x_1}^{x_2} f'(x) dx \right|$$

$$\leq \int_{x_1}^{x_2} |f'(x)| dx \leq \int_0^1 |f'(x)| dx$$

考法七. 轻化为二重积分. 并应用轮换对称性 (以后讲).